

```
# Install once if required:  
# install.packages("reshape2")
```

```
library(reshape2)
```

```
data <- airquality
```

```
cat("Airquality Dataset:\n")  
print(data)
```

```
# Remove missing values  
data <- na.omit(data)
```

```
# Melt data  
melted_data <- melt(  
  data,  
  id.vars = c("Month", "Day")  
)
```

```
cat("\nMelted Data:\n")  
print(head(melted_data, 10))
```

```
# Monthly averages  
monthly_avg <- dcast(  
  melted_data,  
  Month ~ variable,
```

```
fun.aggregate = mean
```

```
)
```

```
cat("\nMonthly Averages:\n")
```

```
print(monthly_avg)
```

The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for data manipulation using `dplyr` and `reshape2` packages. The code includes steps for removing missing values, melting the data, and calculating monthly averages.
- Environment:** Lists the objects in the workspace: `data` (111 obs. of 6 variables), `melted_data` (444 obs. of 4 variables), and `monthly_avg` (5 obs. of 5 variables).
- Console:** Displays the output of the code, including the structure of the melted data and the monthly averages table.

```
11 # Remove missing values
12 data <- na.omit(data)
13
14 # Melt data
15 melted_data <- melt(
16   data,
17   id.vars = c("Month", "Day")
18 )
19
20 cat("\nMelted Data:\n")
21 print(head(melted_data, 10))
22
23 # Monthly averages
24 monthly_avg <- dcast(
25   melted_data,
26   Month ~ variable,
27   fun.aggregate = mean
28 )
29
30 cat("\nMonthly Averages:\n")
31 print(monthly_avg)
32
```

Melted Data:

Month	Day	variable	value
1	5	Ozone	41
2	5	Ozone	36
3	5	Ozone	12
4	5	Ozone	18
5	5	Ozone	23
6	5	Ozone	19
7	5	Ozone	8
8	5	Ozone	16
9	5	Ozone	11
10	5	Ozone	14

Monthly Averages:

Month	Ozone	Solar.R	Wind	Temp
1	5 24.12500	182.0417	11.504167	66.45833
2	6 29.44444	184.2222	12.177778	78.22222
3	7 59.11538	216.4231	8.523077	83.88462
4	8 60.00000	173.0870	8.860870	83.69565
5	9 31.44828	168.2069	10.075862	76.89655